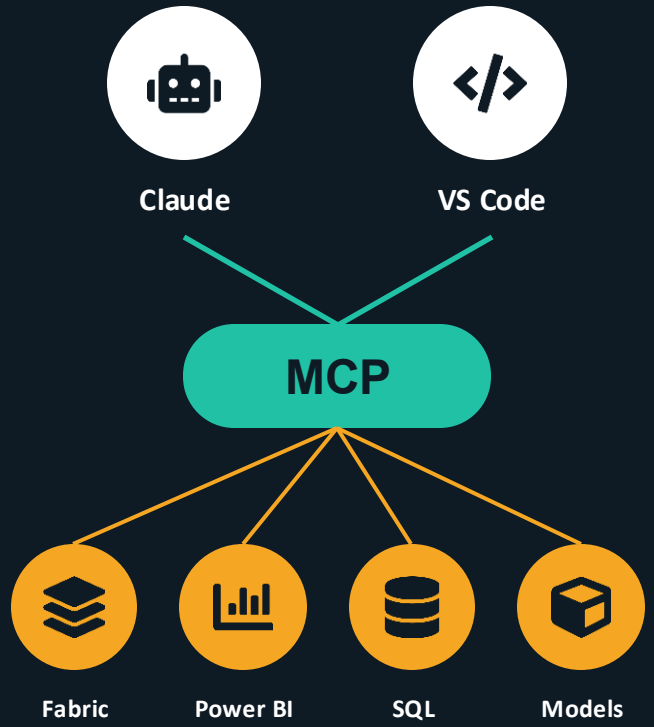
SQL SATURDAY MINNESOTA #1153

Supercharging Fabric with MCP

Connecting Claude and VS Code up to Fabric, Power BI, SQL and semantic models.

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October 24, 2026 | St Paul College



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THE PLAN

How this hour works

10:15

Welcome

5 min

10:20

MCP in 10 minutes

10 min

10:30

Demo 1: Claude

Fabric and Power BI, 15 min

10:45

Demo 2: VSCode

Fabric, SQL, semantic models, 20 min

11:05

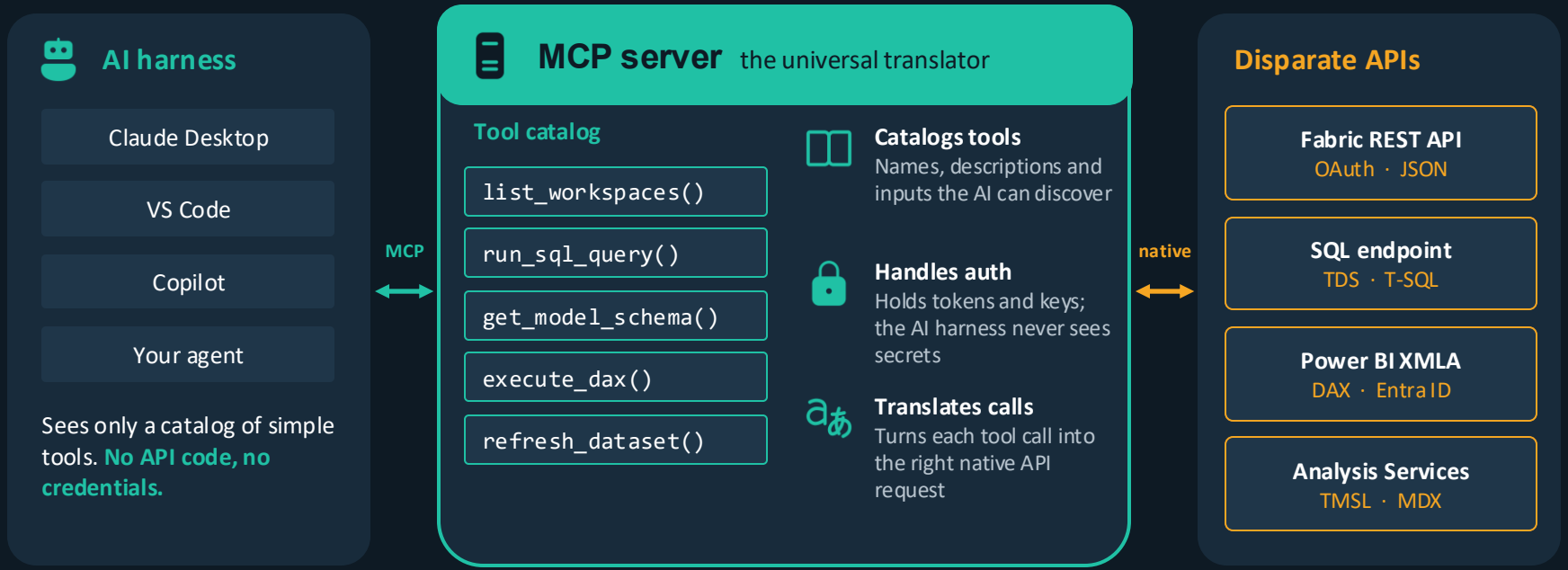
Q and A

10 min

Github: https://github.com/JoelMiller74/sqlsaturday-2026/

MCP IN 10 MINUTES

What is an MCP Server?



An MCP server wraps many different APIs and publishes them as one catalog of simple, self-describing tools. The AI harness just picks a tool; the server handles auth and speaks each API's native language.

What value does an MCP Server provide?

Without MCP: you describe it

Developer types context by hand

"Our orders table has about 12 columns, I think CustomerID is the key… Why did last night's load fail?"

Generic answer
Built on a guess: stale schema, no logs, no tickets. The AI can talk, but it can't look or act.

With MCP: the AI connects to it

Azure
Read logs, inspect resources, check run history

Jira & ticketing
Read ticket details, create and update tickets

Databases & semantic models
Read real schemas, run queries and DAX

GitHub
Start a pull request, review a PR

Microsoft Learn
Search the official documentation

Confluence
Read and update team documents

Real context + real tools: the AI harness works from live logs, schemas, tickets and docs, and can act on them with your permissions, instead of guessing from a description you typed.

MCP IN 10 MINUTES

How it helps data teams

Works with your real data

The AI Harness reads your actual schemas and models, so you don't have to paste them into a chat. It can read server logs, get Fabric item configurations and security, and a whole lot more.

One question, several systems

Ask a question that needs both SQL and Power BI. The AI Harness connects to each relevant MCP Server and tool, gets responses, and gives you a single answer.

Works in any AI Harness

The same MCP server can work in Claude Desktop, VS Code, Gemini or your own agent.

Faster dev work

It can draft DAX, document measures, create or update tickets, and set up Fabric notebooks. You approve each step.

What MCP won't do: copy your data somewhere new or get around your security.

MCP IN 10 MINUTES

How it works

1 · TODAY'S APIS

Messy & non-standard

- REST / JSON
- SOAP / XML
- GraphQL
- CLI / PowerShell
- SQL / ODBC
- Admin portals

2 · MCP SERVERS TRANSLATE

One catalog of named tools

- `query_sql` — READ
  Run a read-only query on a database
- `get_fabric_item` — READ
  Item config, security & lineage
- `update_ticket` — WRITE
  Change status, assign, add a comment

3 · AI HARNESS

Finds, calls, reasons

Claude or GitHub Copilot picks tools by name, description & metadata

- Get data from source systems
- Manage systems & configs
- Make changes (you approve)

No copy & paste
Context flows straight from the source

No swivel-chair
No hopping between consoles

Better decisions
Data from many systems in one answer

Inside the AI Harness, the assistant (Claude or GitHub Copilot) reads tool descriptions to decide what to call.

MCP IN 10 MINUTES

Local MCP vs. Remote MCP

LOCAL Runs where you manage it		REMOTE Hosted endpoint over HTTPS
Your machine, started by the host	Runs on	A hosted endpoint, e.g. mcp.fabric.microsoft.com
Started by the app on your machine	Connects by	A URL over HTTPS
Same: Inherited by the target system (OAuth, etc.)	Sign in	**Same:** Inherited by the target system (OAuth, etc.)
Sometimes requires additional software installs	Setup	**Paste a URL, sign in** ✓
Local files, local services, or remote services ✓	Can reach	Usually remote services
Developers, modeling, prototyping (harder setup)	Best for	Business users, shared access (easier setup)
Usually easier to authenticate ✓	Authentication	Usually harder to authenticate through a AI harness

Outlined with a check = the side with the edge for that row.

One is not better than another. They both have Pros and Cons. It will depend on the user, situation, and tools available.

MCP IN 10 MINUTES

Fabric MCP Servers: Remote vs. Local

REMOTE
Hosted by Microsoft

9

Fabric Core
Workspaces, items, folders & roles

Data Warehouse
Run T-SQL on a warehouse or SQL endpoint

Eventhouse
Discover schemas, query with KQL

Activator
Monitoring rules with email / Teams actions

Operations Agent
Instructions, playbooks & monitoring

Fabric IQ
Find reports & models, run DAX

Power BI Authoring
Author or query semantic models in Fabric

Fabric Data Agent
Ask a published data agent questions

Ontology
Business entities, relationships & definitions

LOCAL
You run & manage it

5

Fabric MCP Server
API docs, OneLake, Core & Factory ops

Data Engineering Notebooks
VS Code: context for Claude Code & Codex

DataFactory
Connections, dataflows, pipelines, Airflow

Real-Time Intelligence
Eventhouse, Eventstreams, Activator, ADX

Power BI Authoring
Desktop, PBIP files or Fabric workspaces

Dashed = same server: Power BI Authoring is one server, remote or local. **Self-hosted samples:** Fabric API for GraphQL, SQL MCP Server (DAB)

Source: Microsoft Learn, "Choose a Microsoft Fabric MCP server" (updated 10/21/2026)

MCP IN 10 MINUTES

Guardrails

RISK	WHAT TO DO
Malicious/Unintended instructions or code Text in a table or doc can tell the assistant what to do	✓ Read each action before you approve it. Never auto-approve changes
MCP Servers you can't trust An MCP server can claim one thing and do another	✓ Use Microsoft or known publishers, on a fixed version
Too much access The assistant can do anything your account can do	✓ Use a limited account and a dev workspace. Start read only
Too many tools switched on More can go wrong, and the wrong tool gets picked	✓ Turn on only what the task needs
Passwords in config files Keys and connection strings end up in git	✓ Use environment variables or VS Code prompts

Admins can also control MCP: Power BI tenant setting, GitHub Copilot policy, Claude organization connectors.

Will my data train the model?

Client and plan	Used for training?	Notes
Claude Free, Pro, Max	Yes, unless you opt out	Feedback you submit, chats flagged for safety review, may be used even when training is off.
Claude Team, Enterprise, API	No	Feedback you submit, chats flagged for safety review, may be used even when training is off. Can be disabled
GitHub Copilot Free, Pro, Pro+	Yes, unless you opt out	Since April 24, 2026. Opt out under Settings > "Privacy"; earlier opt-outs are kept
GitHub Copilot Business, Enterprise	No	Not affected; enterprise-owned repos excluded too. Plus admin policy for MCP server use



Copilot may use
Inputs, outputs, code snippets, cursor context, file names, repo structure, feedback



Copilot won't use
Business/Enterprise data, opted-out users, private repos at rest



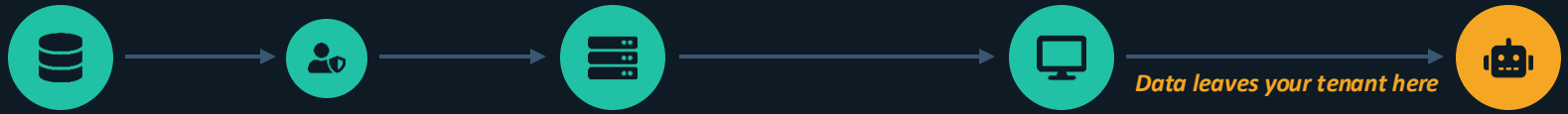
Who gets it
GitHub affiliates incl. Microsoft. Not third-party AI model providers



Worth knowing: the MCP server itself doesn't send your data anywhere. The assistant does, as part of the chat. So what matters is the plan you're signed in with. For work data, use a work account.

MCP IN 10 MINUTES

Where your data goes, and how to control it



Source
Fabric, SQL, Power BI

Your sign-in
Permissions apply

MCP server
Only what its tools allow

AI harness
Shows every tool call

Model
Your plan's terms

LIMIT WHAT'S SHARED

Share less
Use certified models and views, not raw tables. Power BI RLS skips Admins, Members, Contributors

Turn on only the tools you need
Fewer tools = less risk and a more focused agent. Start read-only; add write tools as needed

Use your work plan
Use the company-approved plan: not used for training, but still sent to the provider

Stick to approved servers
Use the MCP servers IT has approved; don't hand your credentials to unvetted servers

VERIFY WHAT HAPPENED

Review before you approve
Read each tool call before approving, especially writes, deletes and admin changes

Check the logs
Your activity lands in Purview audit and Entra sign-in logs. Neither shows what the model saw

Local isn't private
A local MCP server runs on your machine, but whatever it returns is still sent to a cloud model

Connect a server: Claude Desktop vs. VS Code

Claude Desktop

- Remote (HTTP): Connectors > Add custom connector
- Local (stdio): edit the file below, then fully quit (☰ > **File > Exit**)
- Approve tool calls: once, session, or always

`claude_desktop_config.json`

```
"mcpServers": {
  "fabric": {
    "command": "npx", "args": ["-y", "<pkg>"]
  } }
```

VS Code + GitHub Copilot

- Install from the MCP gallery, or define in .mcp.json
- Local (stdio) and remote (HTTP) both go in .mcp.json
- Chat in Agent mode; pick tools in Configure Tools
- Approve tool calls: once, session, workspace, always

`Repository .mcp.json (or user settings)`

```
"servers": {
  "fabric": { "type": "stdio",
    "command": "npx", "args": ["-y", "<pkg>"]
  } }
```

Same server, same settings. Only the top-level key and file differ (**mcpServers** vs **servers**). Both apps run local (stdio) and remote (HTTP) servers; Claude adds remote ones as Connectors.

BEFORE WE START

Ready to follow along

Demo 1: Claude

- Claude Desktop, paid plan vs personal
- Fabric
- Power BI model

Demo 2: VS Code

- VS Code + GitHub Copilot, Agent mode
- Fabric MCP Server extension
- Power BI Modeling MCP extension
- Node.js LTS, .NET 8, Data API builder
- SQL database with AdventureWorksLT

Build along or just watch. Either is fine. If something didn't install, watch today and set it up later from the guide.

DEMO 1A Claude

Claude Desktop to Fabric Core

Local MCP, nothing to install

1. **Add config**
   Settings > Developer > edit config
2. **Start MCP**
   Make sure MCP is running
3. **Sign in**
   Entra ID work account
4. **Enable**
   New chat, approve usage of MCP tools

```
"fabric-mcp-server": {
  "command": "npx",
     "args": [
       "-y",
       "@microsoft/fabric-mcp@latest",
       "server",
       "start",
       "--mode",
       "all"
       ]    }
```

```
# prompts
> List artifacts in a Fabric workspace
> Create a lakehouse
> What analysis will the data enable?
```

DEMO 1B Claude

Claude Desktop to Power BI Desktop

Ask questions of a semantic model (not published – opened in Power BI Desktop)

1. **Add config**
   Settings > Developer > edit config

2. **Open Power BI Desktop Report**
   Pbix file or PBIP file must be open using PBI desktop

3. **Start MCP**
   Make sure Power BI MCP is running

4. **Enable**
   New chat, approve usage of MCP tools

```
"powerbi-modeling-mcp": {
  "command": "npx",
   "args": [
     "-y",
     "@microsoft/powerbi-modeling-mcp@latest",
     "--start"
      ],
      "env": {
        "PBI_MODELING_MCP_ACCEPT_EULA": "true"
       }}
```

```
# prompts
> Write Dax formulas
> Does the model follow best practices
```

DEMO 2 VS Code

VS Code MCP server installation

Install from MCP Catalog, Extensions, or .mcp.json file. .mcp.json file can be user or repository-based.

```
{
    "Fabric Core MCP (Remote)": {
        "type": "http",
        "url": "https://api.fabric.microsoft.com/v1/mcp/core"
    },
    "Fabric Data Warehouse MCP (Global)": {
        "type": "http",
        "url": "https://api.fabric.microsoft.com/v1/mcp/dataPlane/sqlEndpoint"
    },
    "Fabric Eventhouse MCP (Global)": {
        "type": "http",
        "url": "https://api.fabric.microsoft.com/v1/mcp/dataPlane/kqlEndpoint"
    },
    "Fabric IQ MCP (Remote)": {
        "type": "http",
        "url": "https://fabriciq.svc.cloud.microsoft/v1/mcp/fabriciq"
    },
    "Power BI Authoring MCP (Remote)": {
        "type": "http",
        "url": "https://api.fabric.microsoft.com/v1/mcp/powerbi/authoring"
    }
}
```

Remote MCP Servers are built-in HTTP endpoints in Fabric

Local MCP Servers usually require additional configuration, software installation, and steps

Each type of Fabric item has a MCP Server endpoint

Running Local MCP Servers means processing is kept local but data is accessed remotely

DEMO 2 VS Code

Query Data in Fabric

Use MCP Tools to ask questions about Fabric Warehouses, Databases, Semantic Models, and more

The dwh_sqlsaturday warehouse contains a taxi-trip analytics dataset centered on 10,000 trips.

- Trip facts: 10,000 records, 25,000 passengers in total, average trip distance 7.99 miles, average duration 34.96 minutes, and total recorded trip amount $285,930 (average $28.59 per trip).
- Payment summary: Credit Card — 3,340 trips / $95,200; No Charge — 3,330 / $95,940; Cash — 3,330 / $94,790.
- Supporting tables: Date (365 rows), Time (500), Geography (50), Medallion (50), HackneyLicense (25), and Weather (365).
- Coverage: Date rows run from January 1 through December 30, 2024. Geography is 50 ZIP codes in one city in New York, USA.
- Weather: 365 rows; average temperature 60.02°F, with recorded temperatures from 30°F to 89.8°F.

The Trip table links to the date, pickup/drop-off times and geographies, vehicle medallion, and hackney license tables. Weather is keyed by date and geography, so the model supports analysis of fares and trip patterns alongside location, time, and weather.

```
# Agent mode prompts

> tell me what items I have in
the fabric workspace
ws_sqlsaturday_2026

> tell me about the data in the
semantic model sm_sqlsaturday

> tell me about the data in the
database fdb_sqlsaturday

> tell me about the data in the
dwh_sqlsaturday warehouse
```

DEMO 2 VS Code

One prompt. Four servers. One answer.

How the agent fans out across Fabric MCP servers, then fuses the results

EXAMPLE PROMPT

"Why did Midwest revenue drop in March — and set us up to catch it next time?"

Fabric Core MCP › list items + lineage
Finds SalesLakehouse and the warehouse + model that depend on it

Fabric Data Warehouse MCP › T-SQL query
Orders by region/day: Midwest has zero rows Mar 12–15

Real-Time Intelligence MCP › KQL query
Pipeline logs: pl_load_orders failed 4 runs in that window

Power BI MCP › DAX query
Revenue measure shows the dip; no data-freshness check exists



ONE COMBINED ANSWER

Cause: missed loads, not lost sales
Backfill notebook drafted for approval
Freshness measure + alert proposed

Each server sees one system. The agent joins their outputs into one root cause and next steps — and pauses before any write.

DEMO 2 VS Code

More one-prompt ideas

Each prompt chains several Fabric MCP servers into one deliverable



"Is our Gold layer trustworthy this morning?"

SERVERS USED

Fabric Core › run history
Warehouse › row counts, nulls
Power BI › refresh status

Result: one red/amber/green data-health scorecard.



"Onboard the new Returns table end to end."

SERVERS USED

Fabric Core › notebook + pipeline
Warehouse › create vw_Returns
Power BI › table, measures, docs

Result: loaded, modeled and documented in one pass.



"Why are store orders spiking right now?"

SERVERS USED

Real-Time Intel. › KQL on stream
Warehouse › 90-day baseline
Power BI › DAX by store

Result: spike explained vs. baseline, plus an alert rule.

Questions

Three habits to take home

Start read only
Grant writes after you trust the flow

Work account, commercial terms
Know what your plan does with data

Scope and describe
Views, few tools, clear descriptions



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Resources

Lab repo
https://github.com/JoelMiller74/sqlsaturday-2026/

GitHub – Microsoft MCP Servers
https://github.com/microsoft/mcp/tree/main

Choose a Microsoft Fabric MCP Server
https://learn.microsoft.com/en-us/rest/api/fabric/articles/mcp-servers/fabric-mcp-servers-list

Power BI MCP Server Documentation
https://learn.microsoft.com/en-us/power-bi/developer/mcp/

MCP specification
https://modelcontextprotocol.io

SQL MCP Server
https://learn.microsoft.com/en-us/sql/mcp/

MCP Registry – GitHub
https://github.com/modelcontextprotocol/registry

MCP Registry
https://registry.modelcontextprotocol.io/

VS Code – MCP configuration reference
https://code.visualstudio.com/docs/agents/reference/mcp-configuration

VS Code – Manage MCP servers
https://code.visualstudio.com/docs/agents/run/manage-mcp-servers

Claude Code – MCP quickstart
https://code.claude.com/docs/en/mcp-quickstart

Feedback

